

Adaptive Bandit Algorithms for Non-Stationary Recommendation Systems

Evaluating Algorithm Performance on Drifting User Preferences
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Project Description and Outline

- User interests evolve due to trends, context, and long-term preference shifts
- Comparing the efficacy of adaptive non-linear bandit algorithms for accommodating user drift for dynamic recommender systems
- Reproducible synthetic environments allow controlled comparison of adaptive algorithms

- User drift occurs independently of system actions, creating non-stationary reward distributions
- Classical bandits (UCB, ϵ -greedy) assume stationary rewards and over-weight stale data
- Non-stationary bandits explicitly model changing reward distributions
- This project evaluates four adaptive bandit algorithms on synthetic drifting datasets

Synthetic User Environment

- Users are initialized with near-uniform preferences across arms (no initial bias)
- At each timestep, the latent preference vector is updated and renormalized
- Each user has a latent preference vector θ_t over arms
- Arms correspond to content categories or items in a recommender

- Sliding-Window UCB (SW-UCB).
- Non-Stationary ε -Greedy.
- EXP3.S.
- CUSUM-UCB.

Sliding-Window UCB (SW-UCB)

Table: Comparison of UCB and SW-UCB Algorithms

Feature	UCB	SW-UCB
Reward estimation	Uses all past observations to calculate empirical mean	Uses only most recent τ rounds for estimation
Adaptation	Slow (equal weighting of all data)	Rapid (discounts outdated data)

Sliding-Window UCB (SW-UCB)

Arm Selection Rule:

$$I_t = \begin{cases} t & \text{for } t = 1 \text{ to } K \\ \arg \max_{1 \leq i \leq K} (\bar{X}_t(\tau, i) + c_t(\tau, i)) & \text{for } t = K + 1 \text{ to } T \end{cases}$$

Update Rule:

$$N_{t+1}(\gamma, i) = \sum_{s=t-\tau+2}^{t+1} \gamma^{t+1-s} \mathbf{1}_{\{I_s=i\}}$$

$$\bar{X}_{t+1}(\tau, a) = \frac{1}{N_{t+1}(\tau, a)} \sum_{s=t-\tau+2}^{t+1} X_s(i) \mathbf{1}_{\{I_s=i\}}$$

$$c_{t+1}(\tau, i) = B \sqrt{\frac{\xi \log((t+1) \wedge \tau)}{N_{t+1}(\tau, i)}}$$

Variable Notes:

- τ = window size (recent rounds)
- $\gamma \in (0, 1)$ = discount factor
- B = upper bound on rewards
- $\xi > 1/2$ = exploration constant

Non-Stationary Epsilon-Greedy

Arm Selection Rule:

$$I_t = \begin{cases} t & \text{for } t = 1 \text{ to } K \\ \text{random arm in } \{1, \dots, K\} & \text{with probability } \varepsilon \\ \arg \max_{1 \leq i \leq K} Q_t(i) & \text{with probability } 1 - \varepsilon \end{cases} \quad \text{for } t = K+1 \text{ to } T$$

Update Rule:

$$Q_{t+1}(i) = Q_t(i) + \alpha (R_t - Q_t(i)) 1_{\{I_t=i\}}$$

Exponential Recency-Weighted Form:

$$Q_{t+1}(i) = (1 - \alpha)^t Q_1(i) + \sum_{s=1}^t \alpha (1 - \alpha)^{t-s} R_s 1_{\{I_s=i\}}$$

Variable Notes:

- $\alpha \in (0, 1]$ = step-size parameter (larger α = faster adaptation)
- ε = exploration probability
- R_t = reward at time t
- $(1 - \alpha)$ = decay factor for exponential weighting

Arm Selection Rule:

$$p_i(t) = (1 - \gamma) \frac{w_i(t)}{\sum_{j=1}^K w_j(t)} + \frac{\gamma}{K}, \quad i = 1, \dots, K$$

Draw action I_t according to distribution $p(t)$.

Reward Estimation:

$$\hat{x}_j(t) = \begin{cases} \frac{x_j(t)}{p_j(t)} & j = I_t \\ 0 & j \neq I_t \end{cases}$$

Weight Update Rule:

$$w_j(t+1) = w_j(t) \exp\left(\frac{\gamma}{K} \hat{x}_j(t)\right) + \frac{e\alpha}{K} \sum_{i=1}^K w_i(t)$$

Variable Notes:

- γ = exploration rate
- $w_i(t)$ = exponential weight for arm i
- α = stabilization parameter (enables tracking of switching optimal arms)
- Additive term enables regret bounds based on number of switches

Arm Selection Rule:

$$I_t = \begin{cases} \arg \max_{1 \leq i \leq K} (\bar{X}_t(i) + C_t(i)) & \text{with probability } 1 - \alpha \\ \text{uniformly random arm in } \{1, \dots, K\} & \text{with probability } \alpha/K \end{cases}$$

CUSUM Change Detection:

$$g_k^+ = \max\{0, g_{k-1}^+ + s_k^+\}, \quad g_k^- = \max\{0, g_{k-1}^- + s_k^-\}$$

where $s_k^\pm = (y_k - \hat{u}_0 \mp \varepsilon) \mathbf{1}_{\{k > M\}}$.

Restart arm when $g_k^+ \geq h$ or $g_k^- \geq h$.

Update Rule (Post-Restart):

$$N_t(i) = \sum_{s=\tau_i}^t \mathbf{1}_{\{I_s=i\}}, \quad \bar{X}_t(i) = \frac{1}{N_t(i)} \sum_{s=\tau_i}^t X_s(i) \mathbf{1}_{\{I_s=i\}}$$

$$C_t(i) = \sqrt{\frac{\xi \log n_t}{N_t(i)}}, \quad n_t = \sum_{j=1}^K N_t(j)$$

Variable Notes: τ_i = last restart time for arm i ; h = CUSUM threshold; $\hat{u}_0$ = pre-change mean estimate; M = initialization window size.

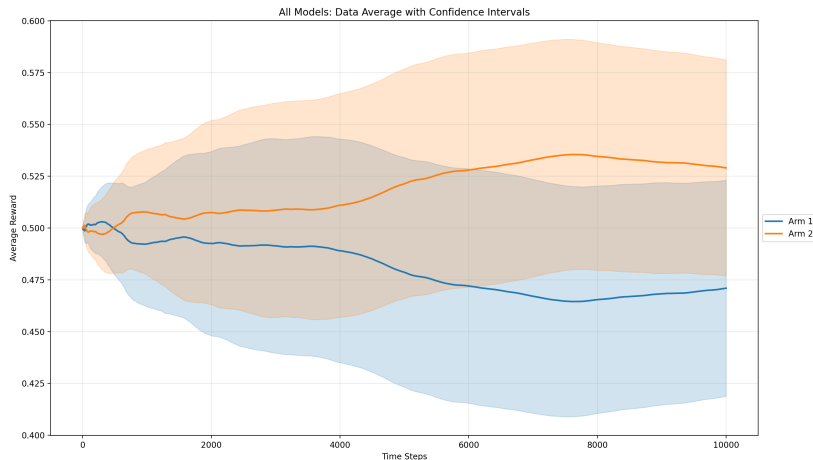


Figure: Mean rolling average reward over $T = 10,000$ rounds.

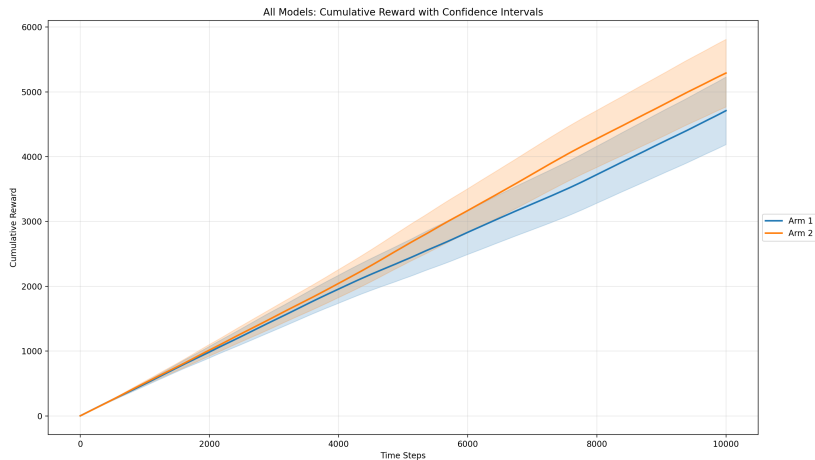


Figure: Average and cumulative reward performance for all models.

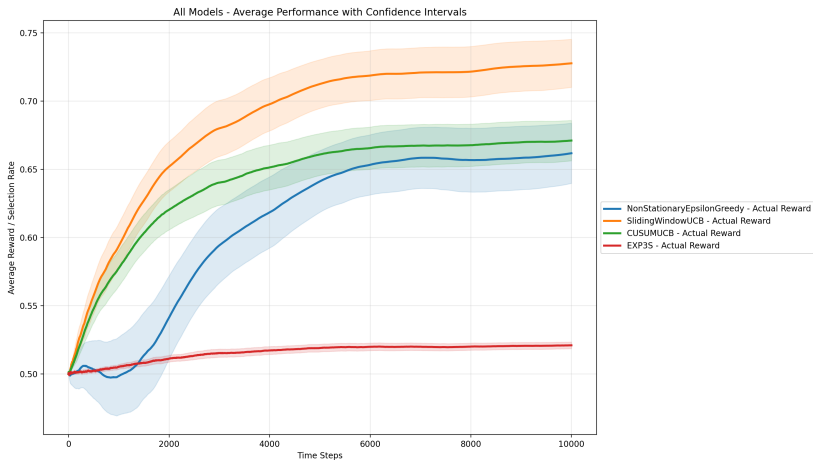


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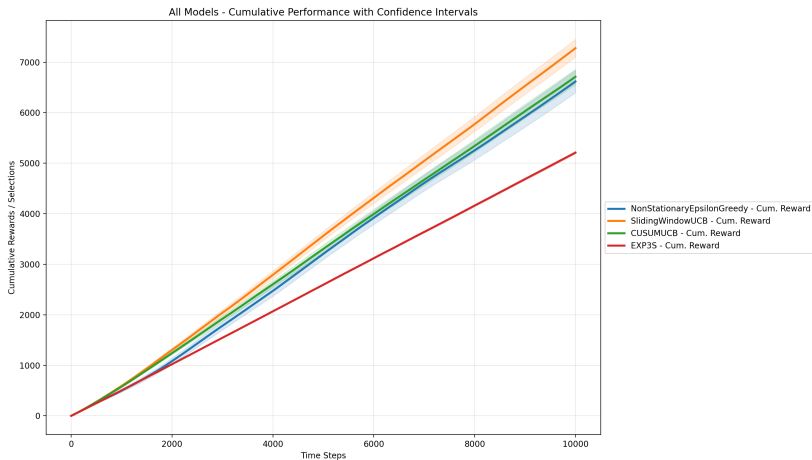


Figure: Average and cumulative reward performance for all models.

- SW-UCB focuses only on recent τ rounds - forgets outdated data
- The rolling average helps it in quick adaptation when user tastes shift
- Arms are drawn from a probability distribution for EXP3.S rather than e-soft methods as of other algorithms
- Conservative exploration wastes reward opportunities for EXP3.S

Thank You

Questions?

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